ANSHIKA KESHARI

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Education

Kalinga Institute of Industrial Technology (KIIT), Bhubaneswar, Odisha — *2023 – 2027*
Bachelor of Technology in Computer Science & Engineering
Relevant Coursework: Data Structures & Algorithms, Operating Systems, DBMS, OOP using Java, Machine Learning

Gurukul Public School, Kawardha, Chhattisgarh — *2021 – 2023*
Class 12th (CBSE Senior Secondary – Science Stream) — *Class 10th (CBSE Secondary)*

Research & Professional Experience

Research Intern – Cybersecurity & Cyber Resilience — *Mar 2026 – Present*
Dr. SPM International Institute of Information Technology (IIIT Naya Raipur) — *Raipur, Chhattisgarh*

- **INTPART ResCri Project**: Contributing to the **India–Norway collaborative research project** between IIIT Naya Raipur and Norwegian research institutions on cyber resilience for critical infrastructure using Digital Twins.
- **Graph Digital Twin Architecture**: Engineered a Physics-Informed **Graph Digital Twin (GDT)** framework combining Spatial-Temporal GNNs (**ST-GNN**) and **RGCN** for real-time anomaly detection and IT-OT cyber-attack classification.
- **Physics-Informed ML & Privacy**: Formulated AC Power Flow physics loss constraints (Ohm's & Kirchhoff's laws), achieving **96.8% F1-score** while analyzing privacy preservation techniques in critical infrastructure data.
- **Explainable AI (XAI) & Automation**: Integrated **GNNExplainer** and **Integrated Gradients** for node/edge attribution, powering sub-15ms automated cyber-physical attack mitigation.

AI/ML & Cybersecurity Summer Intern — *May 2026 – Jul 2026*
School of Computer Applications, KIIT Deemed to be University — *Bhubaneswar, Odisha*

- Completed an 8-week intensive summer internship program covering **AI/ML workflows, Cybersecurity, Image Processing, and NLP**.
- Developed computer vision models and NLP classification pipelines, applying core security principles to intelligent data processing.
- Conducted model performance evaluation using Scikit-learn, Pandas, and Matplotlib for feature selection and metric validation.

Key Projects

Anti-Money Laundering (AML) Detection System — *Python, Random Forest, XGBoost, Scikit-learn, Pandas, Git*

- Engineered an end-to-end ML classification pipeline using Random Forest and XGBoost to analyze 10,000+ financial transaction records, achieving **98.2% precision** in classifying high-risk fraudulent transactions.
- Applied feature engineering on 15+ behavioral attributes (transaction velocity, account age, spatial density patterns) and evaluated models using precision, recall, and F1-score metrics.
- Implemented data cleaning, imbalance handling (SMOTE), hyperparameter tuning, and cross-validation for model generalization.
- Established complete project lifecycle management with Git, maintaining modular OOP architecture and clean commit history.

Smart Infrastructure Anomaly & Cyber-Physical Simulator — *Python, PyTorch, Pandapower, NetworkX*

- Built an interactive grid topology simulator evaluating cyber-attack scenarios (False Data Injection & Load Altering Attacks) across IEEE 14-bus, 39-bus, and 118-bus power grid benchmarks.
- Developed real-time telemetry processing pipelines converting power grid bus voltages and phase angles into PyTorch graph data.

Skills & Competencies

- **Languages**: Java, Python, SQL, C/C++ (Basics), HTML/CSS, LaTeX
- **AI / ML & Security**: Scikit-learn, XGBoost, Random Forest, PyTorch (Basics), Pandas, NumPy, OpenCV, NLP, Graph Neural Networks (ST-GNN, RGCN), Physics-Informed ML
- **Tools & Databases**: Git, GitHub, VS Code, Postman, Linux (Ubuntu), MySQL, Relational DBMS
- **CS Fundamentals**: Data Structures & Algorithms, Operating Systems, OOP (Java), Database Management Systems (DBMS)
- **Soft Skills & Attributes**: Problem Solving, Technical Communication, Analytical Thinking, Self-Learning

Certifications & Leadership

Certifications

- **8-Week Summer Internship Certificate in AI/ML, Cybersecurity, Image Processing & NLP** – KIIT — *Jul 2026*
- **Dynamic Programming Camp** – Issued by AlgoUniversity — *May 2026*
- **Corporate Governance** – Coursera | **Corporate Responsibility** – London Business School — *2026*

Positions of Responsibility

- **Model Head, Kzarshion (Krystal Fashion Society)**, KIIT University: Coordinated society events and led 30+ members.
- **Head Girl**, Gurukul Public School (Class 12): Represented 1,000+ students, organized school events, and led student council.

Awards & Achievements

- **1st Position – Melange'26 Fashion Show**, KIITfest 2026 (State-level University Festival).
- **Secured Runner-Up Position** in National Group Fashion Showcase at IIT Guwahati.
- **Completed 6-Year Diploma in Kathak Classical Dance**, demonstrating artistic discipline and performance excellence.